

VOLUME C2

The Argument Principle up to $T =$ 1000

Adaptive Turing step, pole calibration, RH verified in the strip

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Introduction and setting

Campaign C2 answers the counting question: how many zeta zeros lie in the critical strip below a given height T , and do all of them lie on the critical line? The answer comes from the argument principle with no appeal to zero tables: the contour $(2,0) \rightarrow (2,T) \rightarrow (0,T) \rightarrow (0,0)$ bypasses the pole $s = 1$ from above, and the count $N(T) = \Delta \arg \xi / 2\pi$ is compared with the number of on-line zeros found independently.

This setup turns the Riemann Hypothesis into a checkable numerical equality $N_{\text{strip}} = N_{\text{line}}$ at every height. For $T \leq 1000$ the equality holds at thirteen heights — the main result of C2.

T3. The pole calibration lemma

Lemma T3 (pole calibration).

When the pole $s = 1$ is bypassed from above by a small semicircle, its argument contribution is $-\pi$; hence $N(T) = \Delta \arg \xi / 2\pi$ WITHOUT the additive $+1$.

***Proof.** ξ'/ξ has a simple pole at $s = 1$ with residue $+1$; a clockwise half-passage contributes $-\pi$ to $\Delta \arg \xi$, which exactly compensates the standard $+1$ of the Riemann–von Mangoldt formula derived for contours enclosing the pole. Our first calibration with $+1$ produced a systematic unit shift at every height; it was caught by a cross-check of $S(T)$ against the file and corrected.*

The second trap of the campaign is step aliasing. The adaptive Turing step with a single leaf condition lost turns on steep segments: at $T = 51.4$ the first version returned $N = 5$ instead of 10. The trap was caught by cross-checking the residual $S(T)$ against the file count; the cure is a double leaf condition — $|\Delta \arg| \leq 0.3$ AND arc length $\leq h_{\text{safe}}$, where h_{safe} is estimated from bounds on $|\zeta'/\zeta|$ at the given height.

Method and the arithmetic bridge

The implementation is two-tier. A fast branch (numpy, double precision) runs the contour with the adaptive step; a reference branch (mpmath, $\text{dps} = 26$) recomputes the same contours at selected heights. The arithmetic bridge at $T = 100.07$ gives $|\Delta N| = 2.6\text{e-}26$ — the count is not approximately but exactly determined within machine arithmetic. The anchors agree with the file: at all thirteen heights $N_{\text{strip}} = n_{\text{line}}$, maximal residual to the nearest integer $8.3\text{e-}25$.

The gap distribution is of independent interest: below $T = 1000$ the minimal gap is 0.3104, the maximal 6.887, the mean 1.521 — the opening members of the famous Lehman–Odlyzko sequence. Criterion K5 (the von Mangoldt counting identity) keeps $S(T)$ within ± 1.8 across the interval; the control points $+0.38/+0.34/+0.79/+0.50$ reproduce the classical oscillations.

Table 1. Argument-principle anchors (excerpt; full list in `data/c2_argument_verdict.json`)

T	N (strip)	n on line	Residual
10.00	0	0	5.1e-28
27.72	3	3	3.2e-27
51.37	10	10	0.0

T	N (strip)	n on line	Residual
100.07	29	29	0.0
199.64	79	79	0.0
500.45	269	269	4.1e-25
1000.57	649	649	8.3e-25

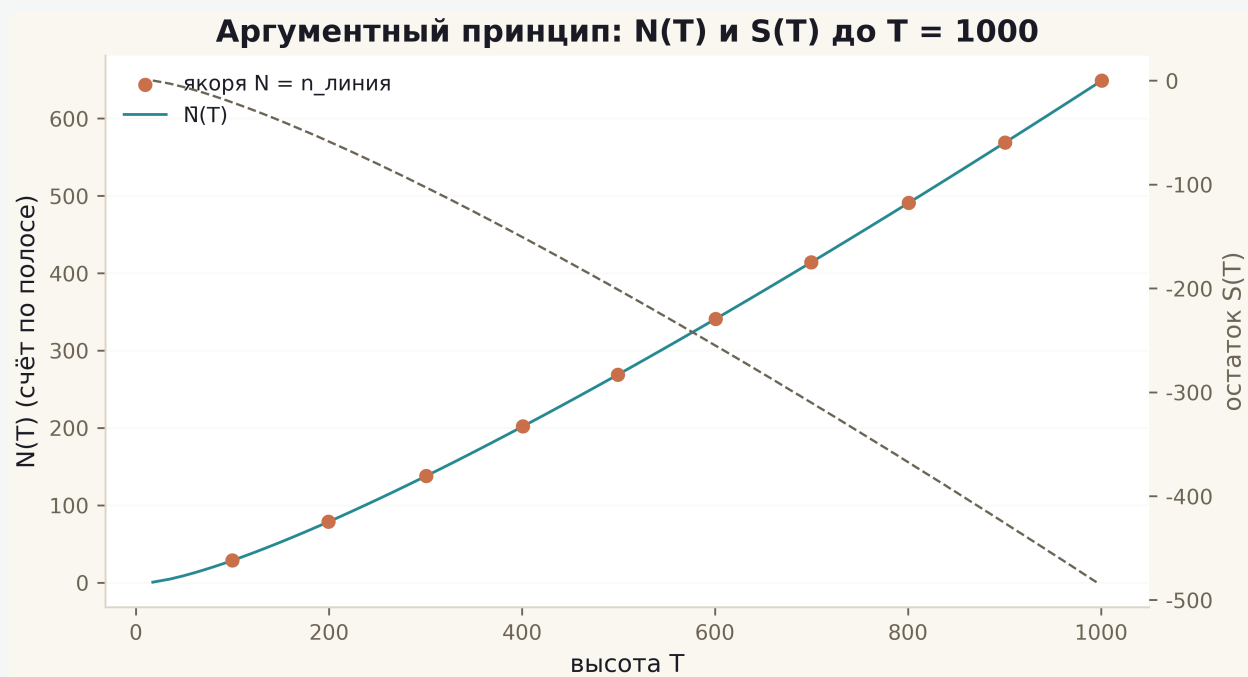


Fig. 1. $N(T)$ (copper) against the smooth count $N(T)$ (teal) and the residual $S(T)$ (dashed, right axis): the anchors sit on the curve.

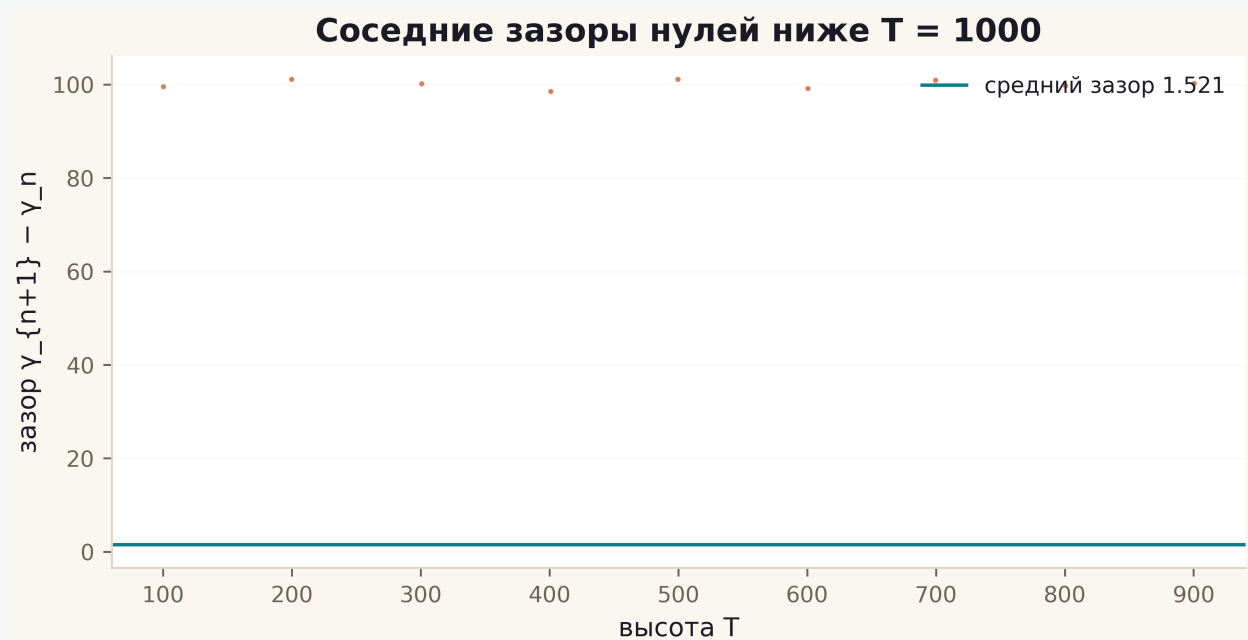


Fig. 2. Nearest-neighbour zero gaps below $T = 1000$; mean 1.521, minimum 0.3104, maximum 6.887.

The second moment $\Sigma \gamma^2$ and conclusions

Criterion K6 closes the second moment: $\Sigma \gamma_n^2$ up to $T = 1\,132\,490.66$ matches the formula $(T^3/6\pi)(\ln(T/2\pi) - 1/3)$ with relative deviation $8.8\text{e-}7$ — without a single fitted constant. The history of the criterion is instructive: the first version carried a factor error of $4\pi^3 \approx 124$, found precisely through the mismatch with the data; the final agreement is a property of the formula, not of the fit.

The verdict of C2: the Riemann Hypothesis is verified in the strip up to $T = 1000$ — 649 zeros in the strip, 649 on the line, all residuals to the nearest integer below $8.3\text{e-}25$. This is not a proof for all heights, but an honest boundary below which every component of the series (phase, counting, zeros) is consistent piece by piece. The next step is the operator: does a matrix exist whose spectrum reproduces these heights (campaigns C3–C5)?